Incident Report: SYN Flood DoS Attack

Explaining a web server outage at a travel agency

Type	Cybersecurity incident report
Tools	Packet sniffer (Wireshark)
Protocols	TCP (three-way handshake), HTTP
Skills	Recognising network attacks, packet analysis, explaining technical impact to a manager
Source	Google Cybersecurity Professional Certificate, Course 3 (*Connect and Protect: Networks and Network Security*)

Situation

A travel agency's sales website stopped responding. Employees use it to search for holiday packages for customers. Monitoring raised an alert, and visitors got a **connection timeout** error. A packet capture showed a very large number of **TCP SYN requests from one unfamiliar IP address**. The web server couldn't keep up with them. As an immediate response, the server was taken offline so it could recover, and the attacking IP was blocked at the firewall.

Section 1: Type of attack

The most likely cause of the connection timeout is a denial-of-service (DoS) attack, specifically a SYN flood.

The logs show the web server receiving an unusually large volume of TCP SYN packets from **a single source IP** and gradually losing the ability to respond. Because the traffic comes from one source, this is a **DoS** attack, not a distributed (DDoS) attack.

Section 2: How the attack disrupts the website

Normal TCP three-way handshake:

1. **SYN:** The client sends a SYN packet asking to open a connection. The packet carries the client's initial sequence number.
2. **SYN-ACK:** The server reserves resources for the connection and replies with SYN-ACK. This acknowledges the client and sends the server's own sequence number.
3. **ACK:** The client sends a final ACK. The connection is now established and data (such as the HTTP request for the webpage) can flow.

What a SYN flood does:

- The attacker sends SYN packets far faster than normal, often from spoofed source addresses.

- The server answers each one with a SYN-ACK and keeps a **half-open connection** waiting for the final ACK.
- The ACK never comes. The half-open connections pile up until the server's **connection backlog is full**.
- Once the backlog is full, the server can't accept new connections, so legitimate SYN requests are dropped or time out.

What the logs show, and the business impact:

At first, legitimate employee traffic and attack traffic were both being handled. As the flood grew, the server spent its resources on the attacker's half-open connections, so it stopped completing handshakes with employees. Their browsers showed **connection timeout** errors. In practice, employees couldn't open the sales site to find holiday packages, which directly affected customer service and sales until the server was recovered.

Recommendations

Blocking the single IP is a stopgap, because an attacker can easily switch to spoofed or different addresses. Longer-term measures:

Measure	Purpose
Turn on **SYN cookies** on the web server or load balancer	The server doesn't reserve memory until the handshake completes, which removes the backlog exhaustion
Rate-limit SYN packets per source at the firewall	Caps how much traffic any single host can send
Deploy an **IPS / next-generation firewall** with DoS signatures	Detects and drops flood patterns automatically
Put the site behind a **CDN or cloud DDoS-protection service**	Absorbs volumetric attacks before they reach the server
Set **monitoring thresholds** for SYN rate and half-open connections	Earlier alerts and faster response

Reflection

Linking each step of the TCP handshake to a server resource made it clear why a "small" packet can take a server down. It also showed why the fix belongs in the handshake design (SYN cookies) and not only in IP blocking.